Haoling (Brian) Pu

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Artificial Intelligence and Innovation, School of Computer Science. GPA: 4.00/4.00. *May 2027*
Relevant Coursework: AI Engineering, Deep Learning, Multimodal ML, NLP, AI Agents, LLM Systems, Generative AI

University of Michigan - Ann Arbor Ann Arbor, MI
Bachelor of Science in Computer Science & Data Science. GPA: 3.95/4.00. *May 2025*
Relevant Coursework: Database Systems, Data Structures and Algorithms, Machine Learning, Web Systems, Computer Organization

Skills

LLM & Agent Engineering: Claude, Codex, Agentic Workflows, Tool Calling, RAG, Vector Databases (Chroma), LangChain
ML, Systems & Cloud: PyTorch, CUDA, MLflow, vLLM, TensorFlow, Slurm, Kubernetes, Docker, AWS, Megatron-SWIFT
Programming Languages: Python, C++, C#, SQL, JavaScript, Go, Java

Experience

Software Engineering Intern 05/2026 – 08/2026
Google *Los Angeles, CA*
- Engineered an agentic knowledge system that auto-generates and self-maintains a knowledge base for a 20+ engineer team
- Deployed as scheduled distributed jobs, automating source discovery, delta ingestion, Gemini synthesis, semantic auditing, and change delivery across 1,000+ engineering docs
- Productionized for cross-team adoption via a reusable AI agent skill delivering source-cited answers for onboarding and knowledge discovery

Research Assistant 10/2025 – Present
CMU Language Technologies Institute, Li Lab *Pittsburgh, PA*
- Developed a reference-free, future-consistent decoding algorithm for simultaneous speech translation using Gemma and Qwen future predictions, achieving COMET parity with a reference-based baseline
- Engineered a resumable, distributed vLLM/Slurm inference pipeline across up to 24 NVIDIA L40S GPUs for parallel generation of streaming translation trajectories
- Generated training data from 40K GigaSpeech utterances and LoRA-fine-tuned Qwen3-Omni with Megatron-SWIFT, improving +5.4 BLEU on ACL en-zh testset via SimulEval

Backend Development Intern 05/2025 – 08/2025
WeRide *Shanghai, CN*
- Reduced Knative cold-start latency by 88% (12.2s to 1.5s) via a service warming pool with symlink-based resource mapping and Kubernetes state recovery backed by PostgreSQL
- Productionized a containerized REST service with Bazel testing and CI/CD; scaled Prometheus/Grafana monitoring across a 120-GPU cluster, cutting scan time by 60% at 99.9% uptime

Machine Learning Engineer Intern 05/2024 – 08/2024
UMSN Pregnancy App *Ann Arbor, MI*
- Led a 6-person team to launch a HIPAA-compliant AI prenatal care platform serving 20+ expectant mothers
- Integrated a GPT-4o RAG agent with n8n for personalized care summaries; built Redis/JWT authentication reducing login latency by 60% and database load by 70%

Software Development Engineer Intern 05/2023 – 08/2023
AVIAGE SYSTEMS *Shanghai, CN*
- Built a Pandas-based flight-data pipeline for 13 sensor streams and optimized MySQL-backed REST APIs, reducing p95 query latency by 42% and database load by 55%

Projects

Sparse Attention on Blackwell 02/2026 – 04/2026
NVIDIA MLSys 2026 Competition
- Developed a Blackwell-optimized CUDA kernel for DeepSeek-V3 sparse attention using FlashAttention-style IO-aware fusion to reduce KV-cache traffic and accelerate long-context LLM inference
- Passed 23/23 attention workloads at ~57 us, achieving 22-35x speedup over the PyTorch reference on NVIDIA B200

Hybrid Retrieval-Augmented Generation System 01/2026 – 03/2026
CMU Advanced NLP
- Built an end-to-end RAG pipeline converting HTML and PDF sources into FAISS/BM25 indexes with Reciprocal Rank Fusion and local open-source LLM inference for domain-specific factual QA